

MSC FINANCE & BIG DATA

Optimization Methods with Applications to Finance • 2627\_FI\_5\_3\_073\_E

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# Fundamental Concepts in Optimization

Elements of Topology and Convexity Theory

**Fouad Ben Abdelaziz, Ph.D.**

Distinguished Professor

Head of the MSc Artificial Intelligence for Business

SISCAD Department • NEOMA Business School

BEN ABDELAZIZ Fouad – NEOMA ([neoma-bs.com](http://neoma-bs.com)) • [fba@neoma-bs.fr](mailto:fba@neoma-bs.fr)

Session 1 • Tutorial • 180 min

# Roadmap of the session

1.  $\mathbb{R}^n$  and matrix calculus
2. Infinity and infinity: countable and uncountable
3. Metric spaces and distances
4. Topology: balls, open sets, closed sets
5. Sequences, convergence, closure and interior
6. Convex sets
7. Convex functions, epigraph and hypograph
8. Why convexity is the whole game
9. Tutorial problems

## Takeaway

Sessions 2–9 are about *models* and *algorithms*. This session is about the ground they stand on: when does an optimum **exist**, when is it **unique**, and when can a local search be trusted to find a **global** answer.

# Where this session sits

| Ch | Content                                                    | Slot              | Format          |
|----|------------------------------------------------------------|-------------------|-----------------|
| 1  | <b>Fundamentals: topology and convexity</b>                | <b>D1 morning</b> | <b>Tutorial</b> |
| 2  | Linear programming: graphical method, duality, sensitivity | D1 afternoon      | Workshop        |
| 4  | Nonlinear programming: KKT, Lagrangian duality             | D2 morning        | Tutorial        |
| 5  | Gradient and Newton methods                                | D2 afternoon      | Workshop        |
| 6  | Multi-objective programming                                | D3 morning        | Tutorial        |
| 3  | Combinatorial optimization                                 | D3 afternoon      | Workshop        |
| 8  | Utility, stochastic dominance, risk measures               | D4 morning        | Tutorial        |
| 7  | Financial applications                                     | Online sync.      | Lecture         |
| 9  | Metaheuristics in finance                                  | Self-study        | Reading         |
| –  | Project presentations                                      | Online sync.      | Seminar         |

**Assessment.** Cases 30% • Final exam 50% • Group project 20%.

## $\mathbb{R}^n$ and matrix calculus

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# The space we work in

$$\mathbb{R}^n = \{x = (x_1, \dots, x_n) : x_i \in \mathbb{R}\}$$

By convention an element of  $\mathbb{R}^n$  is a **column**:

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad {}^t x = (x_1, \dots, x_n)$$

and transposition is an involution:

$${}^t({}^t x) = x.$$

## Notation used throughout

- ${}^t x$  — transpose, written as a *left* superscript.
- ${}^t xy = \sum_{i=1}^n x_i y_i$  — inner product.
- $A \in \mathbb{R}^{n \times m}$ , entries  $a_{ij}$ .
- $0, 1$  — vectors of zeros and ones.

Every optimization model in this course is a statement about a subset of  $\mathbb{R}^n$  and a function defined on it. Nothing more exotic is needed.

# Matrix calculus — the two operations we need

**Addition.** For  $A = (a_{ij})$  and  $B = (b_{ij})$  of the same size,  $S = A + B$  has

$$s_{ij} = a_{ij} + b_{ij}.$$

**Product.** For  $A \in \mathbb{R}^{n \times m}$  and  $B \in \mathbb{R}^{m \times p}$ ,  $C = A \times B \in \mathbb{R}^{n \times p}$  has

$$c_{ij} = \sum_{k=1}^m a_{ik} b_{kj}.$$

The inner dimensions must agree: this is the only compatibility rule.

## Quadratic forms

For  $Q \in \mathbb{R}^{n \times n}$  symmetric,

$$q(x) = {}^t x Q x = \sum_{i=1}^n \sum_{j=1}^n q_{ij} x_i x_j.$$

This single object carries *portfolio variance* ( $Q = \Sigma$ ,  $x = w$ ) and reappears in Chapters 4, 5 and 7.

## Takeaway

$Q \succeq 0$  (positive semidefinite)  $\iff {}^t x Q x \geq 0$  for all  $x$ . We will show in Section 7 that this is exactly the condition for  $q$  to be *convex*.

# A first look at convex combinations

Given  $x_1, \dots, x_n \in \mathbb{R}^n$  and scalars  $\lambda_1, \dots, \lambda_n$  with

$$\sum_{i=1}^n \lambda_i = 1, \quad \lambda_i \geq 0,$$

the point  $\sum_{i=1}^n \lambda_i x_i$  is a **convex combination** of the  $x_i$ .

## Finance bridge

You have already met this object. A *portfolio* is a vector of weights  $w$  with  $\mathbf{1}^T w = 1$  and  $w \geq 0$ : the set of long-only portfolios is exactly the set of convex combinations of the individual assets. Section 6 will name this set the *simplex*, and Chapter 2 will optimize over it.

# Infinity and infinity

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# Not all infinities are the same size

## Definition 1 (Equinumerous sets)

Two sets  $A$  and  $B$  have the same cardinality, written  $|A| = |B|$ , if there exists a bijection  $A \rightarrow B$ .

A set is **countable** if it is in bijection with  $\mathbb{N}$ .

### $\mathbb{Z}$ is countable

Enumerate by alternating sign:

0, 1, -1, 2, -2, 3, -3, ...

Every integer appears exactly once, so  $|\mathbb{Z}| = |\mathbb{N}|$  — even though  $\mathbb{N} \subsetneq \mathbb{Z}$ .

### $\mathbb{Q}$ is countable

Write  $\mathbb{Q} = \{p/q : (p, q) \in \mathbb{Z} \times \mathbb{Z}^*\}$  and place the fractions on a grid indexed by  $(p, q)$ . Sweep the grid along diagonals, skipping non-reduced entries. Every rational is reached in finitely many steps.

**Note.** A proper subset can have the *same* cardinality as the whole set: intuition built on finite sets does not survive the passage to infinity.

# $\sqrt{2}$ is not rational

**Proof (by contradiction).** Suppose  $\sqrt{2} = p/q$  with  $p, q$  integers having no common factor. Then

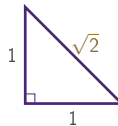
$$\sqrt{2}q = p \implies 2q^2 = p^2.$$

So  $p^2$  is even, hence  $p$  is even: write  $p = 2\beta$ . Substituting,

$$4\beta^2 = 2q^2 \implies q^2 = 2\beta^2,$$

so  $q^2$  is even, hence  $q$  is even.

But then  $p$  and  $q$  share the factor 2 — contradiction.  $\square$



The diagonal of the unit square has a length that no ratio of integers can express. The rationals have *holes*;  $\mathbb{R}$  is what fills them:

$$\mathbb{R} = \{x : x = \lim_{n \rightarrow \infty} x_n, x_n \in \mathbb{Q}\}.$$

# $]0, 1[$ is not countable — Cantor's diagonal

Suppose the reals in  $]0, 1[$  could be listed:

$$\begin{aligned}x_1 &= 0.\underline{x_{11}}\ x_{12}\ x_{13}\ \dots \\x_2 &= 0.x_{21}\ \underline{x_{22}}\ x_{23}\ \dots \\x_3 &= 0.x_{31}\ x_{32}\ \underline{x_{33}}\ \dots \\&\vdots\end{aligned}$$

Construct  $y = 0.y_1y_2y_3\dots$  down the diagonal:

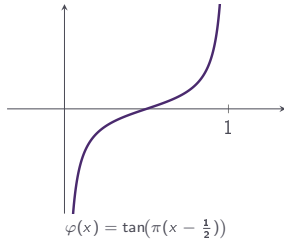
$$y_i = \begin{cases} 9 & \text{if } x_{ii} \neq 9, \\ 1 & \text{if } x_{ii} = 9. \end{cases}$$

Then  $y \in ]0, 1[$  but  $y \neq x_i$  for every  $i$ , since they differ in the  $i$ -th digit. The list was not exhaustive.  $\square$

## Takeaway

$|\mathbb{Q}| < |\mathbb{R}|$ . Moreover  $|\mathbb{R}| = |2^{\mathbb{N}}|$ , and in general  $|2^A| = 2^{|A|}$ : the power set is always strictly bigger than the set.

## $]0, 1[$ has as many points as the whole line



$\varphi : ]0, 1[ \longrightarrow \mathbb{R} = ]-\infty, +\infty[$  is a bijection.

So a bounded open interval and the entire real line are *indistinguishable by cardinality*.

### Extended reals

It is often convenient to adjoin the two endpoints:

$$\overline{\mathbb{R}} = [-\infty, +\infty] = \mathbb{R} \cup \{-\infty, +\infty\}.$$

We use  $\overline{\mathbb{R}}$  when writing  $p^* = +\infty$  for an infeasible problem and  $p^* = -\infty$  for an unbounded one.

# Metric spaces

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# Measuring distance

## Definition 2 (Metric)

A **distance** on  $\mathbb{R}^n$  is a map  $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^+$ ,  $(x, y) \mapsto d(x, y)$ , satisfying

- (i)  $d(x, y) = 0 \iff x = y$ ;
- (ii)  $d(x, y) = d(y, x)$  *(symmetry)*;
- (iii)  $d(x, y) \leq d(x, z) + d(z, y)$  *(triangle inequality)*.

The pair  $(\mathbb{R}^n, d)$  is a **metric space**.

## Takeaway

Everything that follows — balls, open sets, convergence, continuity, and ultimately the existence of an optimal portfolio — is built from these three axioms and nothing else.

# Three distances on $\mathbb{R}^2$

## Euclidean $d_2$

$$d_2(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

The straight line.

## Maximum $d_1$

$$d_1(x, y) = \max(|x_1 - y_1|, |x_2 - y_2|)$$

The worst coordinate.

## Manhattan $d_M$

$$d_M(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

The city grid.

### Worked example

Take  $x = (1, 3)$  and  $y = (6, 2)$ , so  $|x_1 - y_1| = 5$ ,  $|x_2 - y_2| = 1$ . Then

$$d_2 = \sqrt{5^2 + 1^2} = \sqrt{26} \approx 5.1, \quad d_1 = \max(5, 1) = 5, \quad d_M = 5 + 1 = 6.$$

Three numbers for the same pair of points — and, as the next slide shows, three shapes of “ball”.

## Topology: balls, open and closed sets

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# Open balls

## Definition 3 (Open ball)

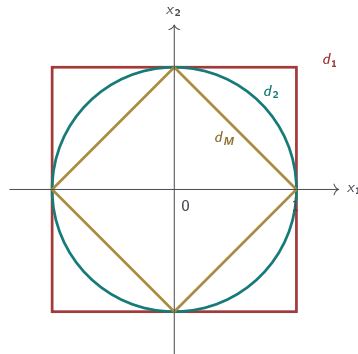
For  $x_0 \in \mathbb{R}^n$  and  $\varepsilon > 0$ ,

$$B_o(x_0, \varepsilon) = \{x : d(x_0, x) < \varepsilon\}.$$

The unit ball  $B_o(0, 1)$  in  $\mathbb{R}^2$  takes three different shapes:

- $d_M$ :  $|x_1| + |x_2| < 1$  — a **diamond**;
- $d_2$ :  $\sqrt{x_1^2 + x_2^2} < 1$  — a **disc**;
- $d_1$ :  $\max(|x_1|, |x_2|) < 1$  — a **square**.

Note the nesting: diamond  $\subset$  disc  $\subset$  square.



# Open sets

## Definition 4 (Open set)

$S \subseteq \mathbb{R}^n$  is **open** if

$$\forall x_0 \in S, \exists \varepsilon > 0 \text{ such that } B_o(x_0, \varepsilon) \subseteq S.$$

### $]0, 1[$ is open

Let  $x_0 \in ]0, 1[$  and take

$$\varepsilon = \frac{1}{2} \min(x_0, 1 - x_0) > 0.$$

Then  $]x_0 - \varepsilon, x_0 + \varepsilon[ \subseteq ]0, 1[$ .

### $[0, 1]$ is not open

Take  $x_0 = 1$ . Every ball  $B_o(1, \varepsilon) = ]1 - \varepsilon, 1 + \varepsilon[$  contains points  $> 1$ , which lie outside  $[0, 1]$ . No  $\varepsilon$  works.

The picture to hold: a set is open when every point has room around it. A point sitting exactly on the edge is what destroys openness.

# The algebra of open sets

## Proposition 1

- (O1)  $\emptyset$  and  $\mathbb{R}^n$  are open.
- (O2) If  $S_1, \dots, S_n$  are open, then  $\bigcap_{i=1}^n S_i$  is open.
- (O3) If  $S_\alpha$  is open for every  $\alpha \in A$  ( $A$  arbitrary), then  $\bigcup_{\alpha \in A} S_\alpha$  is open.

*Proof of (O2).* Let  $x \in \bigcap_i S_i$  and pick  $\varepsilon_i > 0$  with  $B_o(x, \varepsilon_i) \subseteq S_i$ . Set  $\varepsilon = \min_i \varepsilon_i > 0$ ; then  $B_o(x, \varepsilon) \subseteq B_o(x, \varepsilon_i) \subseteq S_i$  for every  $i$ . □

### (O2) needs finiteness

Let  $S_n = ]-\frac{1}{n}, \frac{1}{n}[$ , each open. Then

$$\bigcap_{n=1}^{\infty} S_n = \{0\},$$

which is *not* open. An infinite intersection of open sets can close up.

# Sequences and convergence

## Definition 5 (Convergence)

Let  $(x_n)$  be a sequence in  $S \subseteq \mathbb{R}^n$ . Then  $x_n \rightarrow x$  if

$$\forall \varepsilon > 0, \quad x_n \in B_o(x, \varepsilon) \text{ for all but finitely many } n.$$

## Worked example

Let  $x_n = 1 + \frac{1}{n}$ . Claim:  $x_n \rightarrow 1$ .

Fix  $\varepsilon > 0$ , choose  $N > 1/\varepsilon$ . For  $n > N$ :  $\frac{1}{n} < \varepsilon \Rightarrow 1 < 1 + \frac{1}{n} < 1 + \varepsilon \Rightarrow x_n \in ]1 - \varepsilon, 1 + \varepsilon[$ . □

Every  $x_n > 1$ , yet the limit is exactly 1: *the limit need not resemble the terms*. That is the entire content of the next slide.

# Closed sets

## Definition 6 (Closed set)

$S$  is **closed** if it contains the limit of every convergent sequence of its elements. Equivalently,  $S$  is closed if  $S^c$  is open.

### $]0, 1[$ is not closed

Pick any  $x_0 \in ]0, 1[$  and define

$$x_{n+1} = \frac{1 + x_n}{2}.$$

Every  $x_n$  lies in  $]0, 1[$ , and

$$1 - x_{n+1} = \frac{1}{2}(1 - x_n) \rightarrow 0,$$

so  $x_n \rightarrow 1 \notin ]0, 1[$ .

### $[0, 1]$ is closed

Its complement  $] -\infty, 0[ \cup ]1, +\infty[$  is a union of two open sets, hence open by (O3).

## Proposition 2

(C1)  $\emptyset, \mathbb{R}^n$  closed.

(C2) Finite unions of closed sets are closed.

(C3) Arbitrary intersections of closed sets are closed.

By De Morgan from (O1)–(O3).

# Closure, interior, boundary

## Definition 7

Let  $S \subseteq \mathbb{R}^n$ .

- The **closure**  $\overline{S}$  is the set of limits of all convergent sequences of  $S$  — equivalently, the *smallest* closed set containing  $S$ .
- The **interior**  $\mathring{S}$  is the *largest* open set contained in  $S$  (the union of all open subsets of  $S$ ).
- The **boundary** is  $\text{bdy } S = \overline{S} \setminus \mathring{S}$ .

Always  $\mathring{S} \subseteq S \subseteq \overline{S}$ , and

$$S \text{ open} \iff S = \mathring{S}, \quad S \text{ closed} \iff S = \overline{S}.$$



For  $S = ]0, 1[$ :  $\mathring{S} = ]0, 1[$ ,  $\overline{S} = [0, 1]$ ,  $\text{bdy } S = \{0, 1\}$ .

# Why a finance student should care

## Finance bridge

Maximize expected return over the long-only budget set  $\Delta = \{w \in \mathbb{R}^n : \mathbf{1}^T w = 1, w \geq 0\}$ .

## $\Delta$ is closed and bounded

*Closed*: an intersection of the closed sets  $\{\mathbf{1}^T w = 1\}$  and  $\{w_i \geq 0\}$ , closed by (C3).

*Bounded*:  $0 \leq w_i \leq 1$  for every  $i$ .

## Takeaway

Closed *and* bounded in  $\mathbb{R}^n$  means **compact**. Weierstrass' theorem then guarantees a continuous objective *attains* its maximum on  $\Delta$ : an optimal portfolio **exists**.

## Where it breaks

Drop long-only and allow unlimited leverage: the feasible set is unbounded and  $\max \mathbf{1}^T \mu w = +\infty$ . The model, not the market, has failed — and topology says so before you run a solver.

## Convex sets

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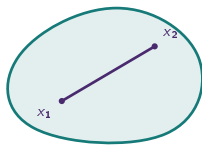
# Convex sets

## Definition 8 (Convex set)

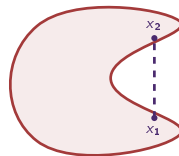
$S \subseteq \mathbb{R}^n$  is **convex** if

$$\forall x_1, x_2 \in S, \forall \lambda \in [0, 1] : \quad \lambda x_1 + (1 - \lambda)x_2 \in S.$$

In words: the whole segment joining any two points of  $S$  stays inside  $S$ .



Convex



Not convex — the segment leaves the set

# Convex hull

## Definition 9 (Convex hull)

Let  $V = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^n$ . Its **convex hull** is

$$\text{conv}(V) = \left\{ \sum_{i=1}^n \lambda_i x_i : \sum_{i=1}^n \lambda_i = 1, \lambda_i \geq 0 \right\}.$$

It is the smallest convex set containing  $V$ . When  $V$  is finite,  $\text{conv}(V)$  is called a **polytope**.

Operations that **preserve** convexity:

- arbitrary intersections  $\bigcap_{\alpha} S_{\alpha}$ ;
- affine images and preimages  $AS + b$ ,  $A^{-1}(S)$ ;
- Cartesian products  $S_1 \times S_2$ ;
- Minkowski sums  $S_1 + S_2$ .

Unions are *not* in this list.

### Finance bridge

The long-only budget set  $\Delta = \{w : \mathbf{1}w = 1, w \geq 0\}$  is exactly  $\text{conv}\{e_1, \dots, e_n\}$  — the convex hull of the “all-in-one-asset” portfolios, whose vertices are the  $n$  single-asset portfolios.

# A catalogue of convex sets in finance

| Set                      | Form                                                        | Convex? |
|--------------------------|-------------------------------------------------------------|---------|
| Budget constraint        | $\{w : \mathbf{1}^T w = 1\}$ (hyperplane)                   | Yes     |
| Return floor             | $\{w : \mathbf{r}^T w \geq \rho\}$ (halfspace)              | Yes     |
| ESG intensity floor      | $\{w : \mathbf{s}^T w \geq \tau\}$ (halfspace)              | Yes     |
| Long-only, position caps | $\{0 \leq w \leq \bar{w}\}$ (box)                           | Yes     |
| Risk budget              | $\{w : \mathbf{r}^T w \Sigma w \leq \sigma^2\}$ (ellipsoid) | Yes     |
| Turnover limit           | $\{w : \ w - w_0\ _1 \leq \kappa\}$ ( $\ell_1$ ball)        | Yes     |
| “Hold at most $K$ names” | $\{w : \ w\ _0 \leq K\}$                                    | No      |
| “Buy 0 or at least 5%”   | semi-continuous variables                                   | No      |

**Read the table this way.** The top block keeps us in Chapters 2, 4 and 5, where problems are solved reliably. The bottom block is precisely what forces us into **branch and bound** (Ch. 3) and **metaheuristics** (Ch. 9).

## Exercise — the cardinality constraint is not convex

Show it in  $\mathbb{R}^2$

Let  $S = \{w \in \mathbb{R}^2 : \mathbf{1}^T w = 1, w \geq 0, \|w\|_0 \leq 1\}$  — “invest in at most one asset”.

Take  $w_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$  and  $w_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ . Both lie in  $S$ .

With  $\lambda = \frac{1}{2}$ :

$$\frac{1}{2}w_1 + \frac{1}{2}w_2 = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}, \quad \|\cdot\|_0 = 2 > 1.$$

The midpoint of two feasible portfolios is infeasible.  $S$  consists of two isolated vertices — it is as far from convex as a set can be.

This one line explains why an index-tracking problem with a “at most 25 holdings” rule is combinatorially hard, while the same problem without it is solved in milliseconds.

# Convex functions

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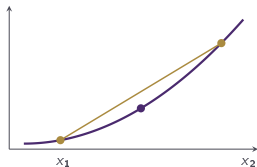
# Convex and concave functions

## Definition 10

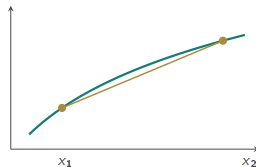
Let  $S \subseteq \mathbb{R}^n$  be convex and  $f : S \rightarrow \mathbb{R}$ . Then  $f$  is **convex** if

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2) \quad \forall x_1, x_2 \in S, \forall \lambda \in [0, 1],$$

and **concave** if the inequality is reversed.



**Convex:** the chord lies *above* the graph



**Concave:** the chord lies *below* the graph

$f$  concave  $\iff -f$  convex. Utility functions are concave (Chapter 8); variance is convex (Chapter 4).

# Epigraph and hypograph

## Definition 11

For  $f : S \rightarrow \mathbb{R}$ ,

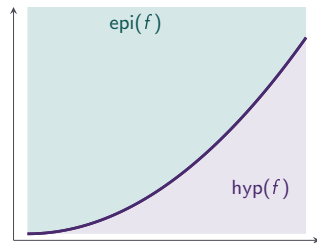
$$\text{epi}(f) = \{(x, y) \in S \times \mathbb{R} : y \geq f(x)\},$$

$$\text{hyp}(f) = \{(x, y) \in S \times \mathbb{R} : y \leq f(x)\}.$$

The epigraph is everything *on or above* the graph; the hypograph everything *on or below*. Their intersection is the graph itself:

$$\text{epi}(f) \cap \text{hyp}(f) = \{(x, y) : y = f(x)\}.$$

**Mind the direction.** “Epi” = above. A sign slip here inverts every theorem that follows.



# The characterisation theorem

## Theorem

Let  $S$  be convex and  $f : S \rightarrow \mathbb{R}$ . Then

$$f \text{ is convex on } S \iff \text{epi}(f) \text{ is a convex subset of } S \times \mathbb{R}.$$

**Proof ( $\Rightarrow$ ).** Let  $(x_1, y_1), (x_2, y_2) \in \text{epi}(f)$ , so  $f(x_1) \leq y_1$  and  $f(x_2) \leq y_2$ . For  $\lambda \in [0, 1]$ , convexity of  $f$  gives

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2) \leq \lambda y_1 + (1 - \lambda)y_2,$$

hence  $\lambda(x_1, y_1) + (1 - \lambda)(x_2, y_2) \in \text{epi}(f)$ .

**Proof ( $\Leftarrow$ ).** Assume  $\text{epi}(f)$  convex. Taking  $y_1 = f(x_1)$  and  $y_2 = f(x_2)$  — both admissible — convexity of  $\text{epi}(f)$  yields  $f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$ . □

## Takeaway

A statement about a *function* becomes a statement about a *set* — letting the geometry of Section 6 do the work of Section 7.

# Building new convex functions from old

## Proposition 3 (Operations preserving convexity)

Let  $f, g$  be convex on a convex set  $S$ . Then:

- (a)  $a_1 f_1 + \dots + a_k f_k$  is convex for any  $a_i \geq 0$ ;
- (b)  $h(x) = \max\{f(x), g(x)\}$  is convex;
- (c) if  $\{f_i\}_{i \in I}$  are convex and bounded above,  $f(x) = \sup_{i \in I} f_i(x)$  is convex;
- (d)  $f(Ax + b)$  is convex for any affine map.

**Proof of (b).** For  $\lambda \in [0, 1]$ ,

$$\begin{aligned} h(\lambda x_1 + (1 - \lambda)x_2) &= \max\{f(\lambda x_1 + (1 - \lambda)x_2), g(\lambda x_1 + (1 - \lambda)x_2)\} \\ &\leq \max\{\lambda f(x_1) + (1 - \lambda)f(x_2), \lambda g(x_1) + (1 - \lambda)g(x_2)\} \\ &\leq \lambda \max\{f(x_1), g(x_1)\} + (1 - \lambda) \max\{f(x_2), g(x_2)\} \\ &= \lambda h(x_1) + (1 - \lambda)h(x_2). \quad \square \end{aligned}$$

# Smooth tests for convexity

## First order ( $f \in C^1$ )

$$f \text{ convex} \iff f(y) \geq f(x) + {}^t\nabla f(x)(y - x)$$

The graph lies above every tangent plane.

## Second order ( $f \in C^2$ )

$$f \text{ convex} \iff \nabla^2 f(x) \succeq 0 \quad \forall x \in S$$

Strictly convex if  $\nabla^2 f \succ 0$ .

## The one that matters

Portfolio variance:

$$f(w) = {}^t w \Sigma w, \quad \nabla^2 f = 2\Sigma.$$

A covariance matrix always satisfies  $\Sigma \succeq 0$ , so **variance is always convex**. If  $\Sigma \succ 0$  it is *strictly* convex, and the minimum-variance portfolio is unique.

Other standard cases: affine functions (convex *and* concave),  $e^x$ ,  $-\log x$ , every norm  $\|\cdot\|$ .

# Audit the machine — is this set convex?

A student asked an AI assistant whether the client's holdings set is convex.

## AI assistant — response

Consider  $S = \{w \in \mathbb{R}^3 : \mathbf{1}^\top w = 1, w \geq 0, \|w\|_0 \leq 2\}$  — long-only portfolios holding at most two of the three assets.

$S$  is **convex**. It is the intersection of three convex sets: the hyperplane  $\{\mathbf{1}^\top w = 1\}$ , the non-negative orthant  $\{w \geq 0\}$ , and the cardinality set  $\{\|w\|_0 \leq 2\}$ . Since an arbitrary intersection of convex sets is convex,  $S$  is convex, and the portfolio problem over  $S$  is therefore a convex program that can be solved reliably.

## Your audit — 15 minutes

1. Is the *conclusion* right or wrong?
2. Which *step* fails? Quote the exact sentence.
3. Give an explicit counterexample: two points of  $S$  whose midpoint is not in  $S$ .
4. The intersection rule it cites is a real theorem. Why does citing a true theorem not rescue the argument?

## Takeaway

The reasoning is fluent, the theorem is real, and the answer is wrong. That combination is the whole reason this exercise exists.

## Why convexity is the whole game

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# Local implies global

## Theorem

Let  $S$  be convex and  $f$  convex on  $S$ . If  $x^*$  is a *local* minimizer of  $f$  on  $S$ , then it is a *global* minimizer.

**Proof.** Suppose not: there is  $\hat{x} \in S$  with  $f(\hat{x}) < f(x^*)$ . For  $\lambda \in ]0, 1]$  put  $x_\lambda = \lambda\hat{x} + (1 - \lambda)x^* \in S$ . By convexity,

$$f(x_\lambda) \leq \lambda f(\hat{x}) + (1 - \lambda)f(x^*) < \lambda f(x^*) + (1 - \lambda)f(x^*) = f(x^*).$$

As  $\lambda \downarrow 0$ ,  $x_\lambda \rightarrow x^*$ , so every neighbourhood of  $x^*$  contains a point with strictly smaller value. That contradicts local optimality.  $\square$

## Takeaway

This is the single most valuable theorem in the course. It is what licenses every algorithm in Chapter 5 — gradient descent, Newton, quasi-Newton — to *stop* when it can no longer improve locally, and report the answer as globally optimal.

# Two more consequences

## Proposition 4 (The solution set is convex)

*If  $f$  is convex on the convex set  $S$ , then  $\arg \min_{x \in S} f(x)$  is convex (possibly empty).*

So a convex problem has either no solution, exactly one, or a whole convex set of them — never two isolated ones.

## Proposition 5 (Uniqueness)

*If in addition  $f$  is strictly convex, the minimizer is unique when it exists.*

**Proof sketch.** If  $x_1 \neq x_2$  were both optimal, the midpoint would satisfy  $f(\frac{x_1+x_2}{2}) < \frac{1}{2}f(x_1) + \frac{1}{2}f(x_2) = p^*$ , contradicting optimality.

## Finance bridge

For the minimum-variance portfolio  $\min_{w \in \Delta} w^T \Sigma w$ :

$\Delta$  compact, variance continuous  $\Rightarrow$  a solution **exists** (Weierstrass)

$\Sigma \succ 0$ , so variance strictly convex  $\Rightarrow$  it is **unique**

variance and  $\Delta$  convex  $\Rightarrow$  local search is **global**

Three facts settle the question before a single number is computed.

# The bridge to this afternoon

When  $f$  is *affine*, the defining inequality holds with equality:

$$f(\lambda x_1 + (1 - \lambda)x_2) = \lambda f(x_1) + (1 - \lambda)f(x_2).$$

An affine function is convex *and* concave.

## Consequence

Optimizing an affine function over a polytope — a **linear program** — always admits an optimal solution at a **vertex** of the polytope.

This is the whole justification for the *corner point method* you will use this afternoon on the carpentry problem:

$$\max 7T + 5C \quad \text{s.t.} \quad 4T + 3C \leq 240, \quad 2T + C \leq 100, \quad T, C \geq 0.$$

Chapter 2 then adds duality and sensitivity analysis, and Case 1 puts them to work.

## Tutorial problems

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# Problems — set A: topology

1. Compute  $d_2$ ,  $d_1$  and  $d_M$  between  $x = (2, -1, 4)$  and  $y = (0, 3, 1)$  in  $\mathbb{R}^3$ . Verify the triangle inequality against  $z = (1, 1, 1)$  for each metric.
2. Draw  $B_o(0, 1)$  in  $\mathbb{R}^2$  for each of the three metrics and prove the nesting  $B_o^{d_M}(0, 1) \subseteq B_o^{d_2}(0, 1) \subseteq B_o^{d_1}(0, 1)$ , where the superscript names the metric used.
3. Show that  $S = \{x \in \mathbb{R}^2 : x_1 > 0, x_2 > 0\}$  is open, and that  $S = \{x \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0\}$  is closed.
4. Let  $S_n = [\frac{1}{n}, \frac{n-1}{n}]$ , each closed. Show  $\bigcup_{n \geq 1} S_n = ]0, 1[$ , which is not closed. Which property fails, and why?
5. Give  $\hat{S}$ ,  $\overline{S}$  and  $\text{bdy } S$  for  $S = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\} \cup \{(1, 0)\}$ .

# Problems — set B: convexity

6. Prove that a halfspace  $\{x : {}^t a x \leq b\}$  is convex, and deduce that every polyhedron  $\{x : Ax \leq b\}$  is convex.
7. Show that  $\{w : {}^t w \Sigma w \leq \sigma^2\}$  is convex when  $\Sigma \succeq 0$ . *Hint: use the epigraph theorem, or convexity of the quadratic form directly.*
8. Is the union of two convex sets convex? Prove it, or give the smallest counterexample you can find.
9. Let  $f(x) = \max\{2x + 1, -x + 4, x^2\}$  on  $\mathbb{R}$ . Prove  $f$  is convex and sketch it. Where is the minimum?
10. The *turnover* constraint is  $\|w - w_0\|_1 \leq \kappa$ . Prove that this set is convex. Then explain, in one sentence, why “turnover in at most 10 positions” is *not*.
11. **Synthesis.** For the problem  $\min_w {}^t w \Sigma w$  s.t.  ${}^t 1 w = 1, {}^t \mu w \geq \rho, w \geq 0$ : state precisely which theorem of today guarantees (a) existence, (b) uniqueness, (c) that a local method suffices. State the assumption each one needs.

# Summary

## Topology gave us

- distances, balls, open and closed sets;
- sequences and limits;
- closed + bounded = compact;
- **existence** of an optimum.

## Convexity gave us

- convex sets, hulls, polytopes;
- convex functions and the epigraph test;
- local = global;
- **uniqueness** and **tractability**.

### Takeaway

Before writing a model ask three questions: is the feasible set *compact*, is it *convex*, is the objective *convex*? Three yeses and the problem is tractable. A no tells you which later chapter you will need.

**Reading.** Full references, with what each source is for, on the next slide.

## References

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# References and further reading

## Core

- Boyd, S. & Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. ch. 2–3: convex sets, convex functions, epigraph, operations preserving convexity.
- Cornuéjols, G., Peña, J. & Tütüncü, R. (2018). *Optimization Methods in Finance*, 2nd ed. Cambridge University Press. ch. 1–2 for the finance framing.

## If you want the proofs in full

- Rudin, W. (1976). *Principles of Mathematical Analysis*, 3rd ed. McGraw-Hill. ch. 2–4: metric spaces, compactness, Weierstrass.
- Rockafellar, R. T. (1970). *Convex Analysis*. Princeton University Press. the definitive account of epigraphs and duality — harder than we need, and where to go next.
- Palomar, D. P. (2024). *Portfolio Optimization: Theory and Application*. Cambridge University Press. ch. 1, for the notation used here.

## Takeaway

**Next.** Chapter 2 — linear programming: graphical method, corner points, duality, sensitivity. Prepare the Excel Solver setup and Case 1.